

Aditya Padhi

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science

Apeejay School

CBSE Board (10th and 12th)

2024 – Present

2022 & 2024

Work Experience

Indian Institute of Technology (IIT) Mandi

AI, Robotics & Research Intern — Drone Lab

May 2026 – Present

- Engineered a unified reward policy for bipedal locomotion generalising across 4 robot URDFs (Berkeley Bolt, Duke Humanoid) with <8% performance drop, eliminating per-robot reward re-tuning entirely.
- Built push-recovery & dynamic disturbance-rejection controllers (PPO/SAC) achieving recovery from 120 N lateral impulses in <1.2 s, cutting fall rate by **63%** vs. PD baselines under randomised perturbation trials.
- Deployed omni-directional locomotion (lin-x, lin-y, ang-z) across MuJoCo, PyBullet, and Isaac Sim/Lab — stable at 1.5 m/s forward · 0.8 m/s lateral · 90°/s yaw, validated on flat and uneven terrain.
- Accelerated policy training **55%** via GPU-parallelised Isaac Lab pipelines (>2048 envs) with domain randomisation over mass, friction, and motor gains across 12 terrain variants, boosting sim-to-real transfer.
- Tech: Python · MuJoCo · PyBullet · Isaac Sim · Isaac Lab · PyTorch · Stable-Baselines3 · PPO/SAC · ROS2 · URDF

Projects

VitaTwin — Real-Time Digital Twin Fatigue System

AI-driven system combining biomechanics, CV, and sequence modeling for fatigue prediction

Tech: Python · MediaPipe · MuJoCo/PyBullet · LSTM/Transformer · OpenCV · Scikit-learn · Streamlit · NumPy

- Built a real-time fatigue detection system using MediaPipe and LSTM/Transformer models, achieving **~85% prediction accuracy**.
- Personalized the system per user using feature normalization and temporal modeling, reducing incorrect fatigue alerts by **~40%**.
- Estimated biomechanical load using physics-based simulation (MuJoCo/PyBullet) with 6+ musculoskeletal parameters.
- Forecasted fatigue trends up to 2 minutes ahead using sequence models (LSTM/Transformer) for proactive intervention.

FocusForge V2 — AI Cognitive Focus System

Real-time system for detecting and optimizing user productivity behavior

Tech: Python · FastAPI · WebSocket · MediaPipe · OpenCV · LSTM · Kalman Filter · SQLite · React 18 · Recharts

- Built a context-switch detection system using MediaPipe and OpenCV, predicting distractions 2–3 seconds early and reducing switches by **32%**.
- Designed a real-time pipeline integrating Kalman filtering and LSTM-based temporal modeling for continuous behavioral analysis.
- Reduced false distraction alerts by **~78%** using Kalman-filtered signal smoothing and probabilistic scoring.
- Improved user refocus time from 42 seconds to under 20 seconds using personalized LSTM-based cognitive modeling.

MediSync AI — Full-Stack Healthcare Platform

AI platform using graph learning and digital twin reasoning for clinical support

Tech: Next.js · FastAPI · PostgreSQL · PyTorch Geometric · NetworkX · SQLAlchemy · Recharts · JWT · TailwindCSS

- Developed a patient similarity engine using Graph Neural Networks (PyTorch Geometric) and cosine similarity, improving matching accuracy by **~35%**.
- Optimized inference pipelines using precomputed embeddings and caching, reducing response time by **~60%**.
- Built a graph-based digital twin model for treatment prediction, achieving **~80% alignment** with historical clinical outcomes.
- Improved system efficiency by **~30%** using modular backend design with FastAPI and SQLAlchemy.

Certifications

Supervised Machine Learning (99.66%) — Nov 2025 — Deep Learning Specialization — 2025 — Agentic AI Fundamentals — Sept 2025 — Introduction to Artificial Intelligence — Jun 2025

Technical Skills

Programming: Python, C++, C, MATLAB, JavaScript

ML & Data Science: Scikit-learn, NumPy, Pandas, Matplotlib

AI & Deep Learning: PyTorch, TensorFlow, Stable-Baselines3, PPO/SAC, LSTM, Transformer, GNN (PyTorch Geometric)

Robotics & Simulation: MuJoCo, PyBullet, Isaac Sim, Isaac Lab, ROS2, MediaPipe, URDF, Domain Randomisation

Backend and Databases: Flask, FastAPI, REST APIs, MongoDB, PostgreSQL, Git/GitHub

Deployment & Cloud: Microsoft Azure, AWS, Google Cloud Platform, Vercel, GitHub Actions, CI/CD Pipelines, Docker, Kubernetes